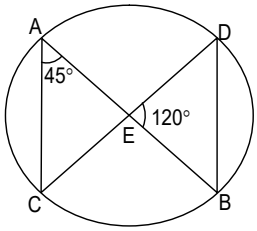
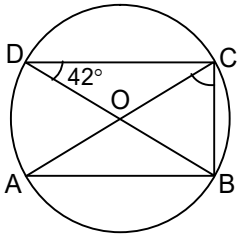
12. In the given figure, two chords AB CD of a circle intersect each other at a point E such that ∠BAC = 45°, ∠BED = 120°. Then find ∠ABD.
1) 15° 2) 30 3) 45° 4) 60°

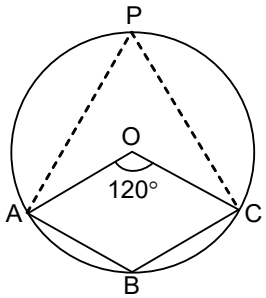


13. In the given circle O is centre and ∠BDC = 42°, then ∠ACB is equal to



1) 60° 2) 48° 3) 45° 4) 42°

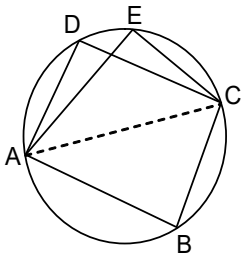
14. The magnitude of ∠ABC, if ∠AOC = 120° will be



1) 125° 2) 120° 3) 130° 4) 135°

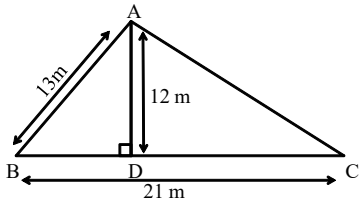
15. A circle is inscribed in a triangle whose sides are 40, 40 and 48 cm respectively. A smaller circle is touching two equal sides of the triangle and to the first circle. Then the radius of smaller circle is
1) 1 2) 2 3) 3 4) 4

16. Given a circle and a quadrilateral ABCD inscribed in it as shown in the figure. If ∠B = 125°, then ∠E

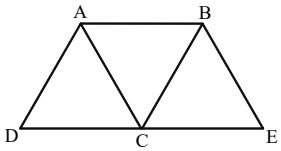


1) 55° 2) 120°
3) 135° 4) 78.5°

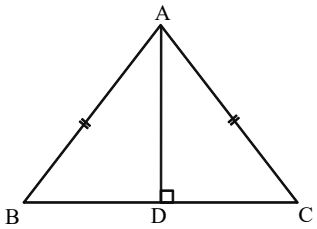
18. In a ΔPQR, if the angles of measures 15°, 10°, and 5° are subtracted from angles P, Q, and R respectively, then the ratio of the angles becomes 4: 5: 6. What is the measure of $\angle$R in ΔPQR?
    1) 55° 2) 60° 3) 65° 4) 70°
19. In the given figure, points B and C represent two persons standing on either side of a 12 m tall building which is represented by AD.



    In the ΔABC so formed, what is the length of side AC?
    1) 16 m 2) 18 m 3) 20 m 4) 22 m
20. If the interior angles of a triangle are in the ratio 5:7:8, then the triangle is
    1) An isosceles triangle 2) An equilateral triangle
    3) A right-angled triangle 4) An acute-angled triangle
21. In a triangle, one of the exterior angles is of measure 105° and the interior opposite angles are in the ratio 3: 4. What is the measure of the greatest interior angle of the triangle?
    1) 45° 2) 65° 3) 75° 4) 95°
22. In ΔABC, AB = AC and $\angle B = \frac{\angle A}{2}$. What type of triangle is ΔABC?
    1) Equilateral triangle 2) Acute-angled triangle
    3) Right-angled triangle 4) Obtuse-angled triangle
23. In the given figure, ABCD and ABEC are two parallelograms.



    If $\angle DAC = 50°$ and $\angle ADC = 70°$, then what is the measure of $\angle CEB$?
    1) 40° 2) 50° 3) 60° 4) 70°
24. In the given figure, AD = 4 cm and BD = 3 cm.



    What is the perimeter of ΔABC?
    1) 16cm 2) 18cm 3) 21cm 4) 22cm

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13. If $\frac{4x-5}{4}-\frac{3x-1}{3}=\frac{13}{12}x$, then the value of x is

1) $-\frac{12}{13}$ 2) $-\frac{10}{13}$ 3) $-\frac{11}{13}$ 4) $-\frac{15}{13}$

14. The number of real solutions of the equation $x^2-3|x|+2=0$ is

1) 2 2) 4 3) 1 4) 3

15. The number of real roots of $\left(x+\frac{1}{x}\right)^3+\left(x+\frac{1}{x}\right)=0$ is

1) 0 2) 2 3) 4 4) 6

16. The number of real roots of $3^{2x^2-7x+7}=9$ is

1) 0 2) 2 3) 1 4) 4

17. $\sqrt{\frac{3x}{x+1}}+\sqrt{\frac{x+1}{3x}}=2, x\neq 0, x\neq -1$ then the solution set is

1) {1} 2) $\left\{\frac{1}{2}\right\}$ 3) {2} 4) $\left\{\frac{1}{3}\right\}$

18. If $|x|^{x^2-2x}=1$, then the values of x are

1) $0, \pm 1, 2$ 2) $0, \pm 1$ 3) $\pm 1, 2$ 4) 1, 2

19. The number of real roots of $\sqrt{x^2-x-6}+\sqrt{6x^2-5x-39}=\sqrt{x^2+4x-21}$ is

1) 1 2) 2 3) 3 4) 4

20. The real roots of the equations $|x^2+4x+3|+2x+5=0$ are

1) $4, -1+\sqrt{3}$ 2) $-4, -1-\sqrt{3}$ 3) $\pm 4, -1\pm\sqrt{3}$ 4) $-2, -4\pm\sqrt{3}$

21. If $6^x+6^y=42$, $x+y=3$ then

1) $x=1, y=2$ 2) $x=-1, y=2$

3) $x=1, y=2$ 4) $x=2, y=1$ or $x=1, y=2$

22. The roots of the equation $\log_2(x^2-4x+5)=x-2$ are

1) 4, 5 2) 2, -3 3) 2, 3 4) 3, 5

23. If the equations $ax+by=1, cx^2+dy^2=1$ have only one solution then $\frac{a^2}{c}+\frac{b^2}{d}=$

1) 1 2) −1 3) 0 4) 2

24. The product of real roots of the equation $|2x+3|^2-3|2x+3|+2=0$, is

1) $\frac{5}{4}$ 2) $\frac{5}{2}$ 3) 5 4) 2

25. $\left\{x\in R:|x-2|=x^2\right\}=$

1) {-1, 2} 2) {1, 2} 3) {-1, -2} $4) {1, -2}

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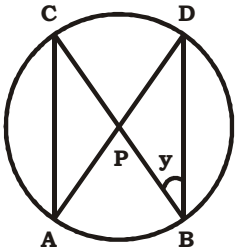
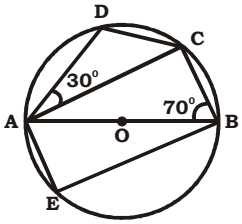
9. In the given figure, if $\angle ACB = 40^0$, $\angle DPB = 120^0$, then y will be



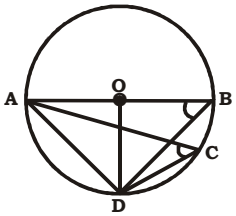
1) 40^0 2) 20^0 3) 0^0 4) 60^0

10. In figure, AB is a diameter of a circle with centre O. If $\angle ABC = 70^0$, $\angle CAD = 30^0$ and $\angle BAE = 60^0$. Then $\angle BAC$ is



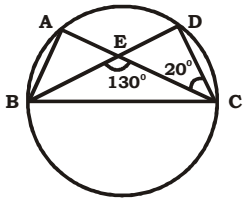
1) 20^0 2) 40^0 3) 110^0 4) none of these

11. In the given figure, AB is a diameter of the circle C and the radius OD is perpendicular to AB. If C is any point on $\overset{\frown}{DB}$, then $\angle BAD$ is



1) 45^0 2) 60^0 3) 20^0 4) none of these

12. In the given figure, A, B, C and D are four points on a circle. AC and BD intersect at a point E such that $\angle BEC = 130^0$ and $\angle ECD = 20^0$, then $\angle BAC$ is



1) 50^0 2) 70^0 3) 130^0 4) 110^0

13. In the given figure, the bisector of $\angle B$ of an isosceles triangle ABC with AB = AC meets the circumcircle of ΔABC at P. If AP and BC produced meet at Q. Then which of the following is true.

18. If the point P(2,1) lies on the line segment joining points A(4,2) and B(8,4),then

1) $AP = \frac{1}{3}AB$ 2) AP = PB 3) $PB = \frac{1}{3}AB$ 4) $AP = \frac{1}{2}AB$

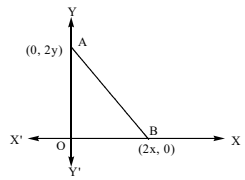
19. If $P\left(\frac{a}{3},4\right)$ is the mid-point of the line segment joining the points Q(-6,5) and R(-2,3),then the value ofa is

1) -4 2) -12 3) 12 4) -6

20. The perpendiclar bisector of the line segment joining the point A(1,5) and B(4,6) cuts the y-axis at.

1) (0,13) 2) (0,-13) 3) (0 ,12) 4) (13, 0)

21. The coordinates of the point which is equidistant from the three vertices of the ΔAOB as shown in the figure is.



1) (x, y) 2) (y, x) 3) $\left(\frac{x}{2},\frac{y}{2}\right)$ 4) $\left(\frac{y}{2},\frac{x}{2}\right)$

22. AOBCis rectangle whose three vertices are A(0,3), O(0,0) and B (5, 0). The length of its diagonal is

1) 5 2) 3 3) $\sqrt{34}$ 4) 4

23. The perimeter of a triangle with vertices (0,4),(0,0) and (3,0) is

1) 5 2) 12 3) 11 4) $7+\sqrt{5}$

24. The area of a triangle with vertices A (3,0), B(7,0) and C(8,4) is

1) 14 2) 28 3) 8 4) 6

25. Find the point on the x-axis which is equidistant from(2,-5) and (-2,9)

1) (-7,0) 2) (-5,0) 3) (7,0) 4) (5,0)

26. Find the value of 'k' for which the point (8,1),(k,-4) and (2,-5) are collinear

1) 3 2) 2 3) 1 4) 0

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CONTENTS

S.NO.	TOPIC	PAGE NO.
1.	Number System	5-16
2.	Surds	17-24
3.	Polynomials	25-35
4.	Coordinate Geometry	36-47
5.	Triangles	48-73
6.	Circles	74-93
7.	Mensuration	94-103
8.	Permutations	103-111
9.	Combinations	112-118
10.	Logarithms	119-127
11.	Solving equations	128-138
12.	Mathematical operations	139-148

10. The remainder when $10^1 + 10^2 + 10^3 + 10^4 + \ldots\ldots\ldots + 10^{99}$ is divided by 6

1) 1 2) 2 3) 0 4) 3

11. Which of the following numbers are of the form 4n(n + 1) + 1

1) 4489 2) 444889 3) 44448889 4) all of these

12. The remainder when an odd perfect square is divided by 8

1) 0 2) 1 3) 2 4) 3

13. For n = 3, $n^3 - n$ represents an _____ number

1) Odd 2) Even 3) Prime 4) Perfect

14. The number of pairs of prime numbers that are present up to 100 with a difference of 10 is

1) 11 2) 2 3) 9 4) 10

15. The value of $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\ldots\ldots\ldots\left(1-\frac{1}{n}\right)$ is

1) 1 2) (n – 1) 3) $\frac{1}{n}$ 4) 0

16. The smallest prime number dividing the sum $3^{11} + 5$ is

1) 2 2) 3 3) 5 4) $3^{11} + 5^{13}$

17. If the unit's digit in the product (47x × 729 × 345 × 343) is 5, what is the maximum number of values that x may take ?

1) 9 2) 3 3) 7 4) 5

18. If 2a = 3b = 7c = 5d = 210 then a + b + c + d is

1) Odd 2) Even 3) Prime 4) Negative

19. The unit digit of $1^1 + 2^2 + 3^3 + \ldots\ldots 10^{10}$ is

1) 0 2) 7 3) 2 4) 4

20. If $n! = \frac{(n+4)!}{(n+1)!}$, then the value of n is :

1) 5 2) 18 3) 6 4) 9

21. The remainder obtained when $23^3 + 31^3$ is divided by 54:

1) 0 2) 1 3) 3 4) Cant be determined

13. Rationalising factor of $\left(27^{\frac{1}{9}} - 4^{\frac{1}{6}}\right)$ is

1) $27^{\frac{2}{9}} - 27^{\frac{1}{9}}.4^{\frac{1}{3}} + 4^{\frac{1}{2}}$ 2). $3^{\frac{2}{3}} + 3^{\frac{1}{3}}.2^{\frac{1}{3}} + 2^{\frac{2}{3}}$ 3) $3^{\frac{2}{3}} - 3^{\frac{1}{3}}.2^{\frac{1}{3}} + 2^{\frac{2}{3}}$ 4) $27^{\frac{2}{9}} + 27^{\frac{1}{9}}.2^{\frac{1}{3}} + 4^{\frac{1}{2}}$

14. If $\sqrt{19 - 4\sqrt{x}} = \sqrt{12} - \sqrt{7}$, then x =

1)84 2)28 3)21 4)14

15. $\sqrt{12\sqrt{5} + 2\sqrt{55}} =$

1) $\left(\sqrt{11}+1\right)\sqrt[4]{5}$ 2) $\sqrt[4]{5}(1+\sqrt{5})$ 3) $\sqrt[4]{5}(\sqrt{11}+\sqrt{5})$ 4) $\sqrt{11}+1$

16. If $\frac{4+3\sqrt{5}}{4-3\sqrt{5}} = a + b\sqrt{5}$, then $(a, b) =$

1) $\left(\frac{61}{29}, \frac{-24}{29}\right)$ 2) $\left(\frac{-61}{29}, \frac{24}{29}\right)$ 3) $\left(\frac{61}{29}, \frac{24}{29}\right)$ 4) $\left(\frac{-61}{29}, \frac{-24}{29}\right)$

17. If $x = \sqrt[3]{9}, y = \sqrt[4]{11}, x = \sqrt[6]{17}$ then:

1) $x > y > z$ 2) $y > z > x$ 3) $z > y > x$ 4) $x < z < y$

18. $\sqrt{\sqrt{50} + \sqrt{48}} = k\left(\sqrt{3} + \sqrt{2}\right)$, then k =

1) $2^{1/2}$ 2)2 3) $2^{1/4}$ 4)0

19. The positive square root of 11 + $\sqrt{112}$ is :

1) $\sqrt{7}+2$ 2) $\sqrt{7}+\sqrt{2}$ 3) $2-\sqrt{7}$ 4)2

Q.NO 20-23

A surd is given by x = $7+4\sqrt{3}$

20. Find the value of $x + \frac{1}{x}$

1) 12 2) 14 3) 16 4) 18

21. Find the value of $x - \frac{1}{x}$

1) $8\sqrt{3}$ 2) $4\sqrt{3}$ 3) $3\sqrt{3}$ 4) $3\sqrt{4}$

22. Find the value of $x^2 + \frac{1}{x^2}$

1) 192 2) 212 3) 194 4) 200

23. Find the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$

1) 1 2) 2 3)3 4)4

24. $\sqrt{20 - \sqrt{20 - .....\infty}} =$

1) 4 2) -20 3) $-\sqrt{20}$ 4) 20

13. If $f(x) = 3x^4 + 7x^3 - 15x^2 + x - 2$, then $f(x - 7)$ is

1) $3x^4 - 77x^3 + 720x^2 - 2876x + 4058$ 2) $3x^4 + 77x^3 + 720x^2 - 2876x - 4058$

3) $3x^4 - 77x^3 - 720x^2 + 2876x + 4058$ 4) None of these

14. If $f(x) = x^3 - 6x^2 + 5x + 8$, then $f(x + 2)$ is

1) $x^3 + 7x + 2$ 2) $x^3 + 7x - 2$ 3) $x^3 - 7x + 2$ 4) $x^3 - 7x - 2$

15. By factor theorem, the factors of $x^4 + 2x^3 - 7x^2 - 8x + 12$ are

1) $(x - 1)(x + 2)(x - 2)(x - 3)$ 2) $(x + 1)(x + 2)(x - 2)(x + 3)$

3) $(x - 1)(x + 2)(x + 2)(x + 3)$ 4) None of these

16. By factor theorem, the factors of $x^4 + 3x^3 - 7x^2 - 27x - 18$ are 1) $(x + 1)(x + 2)(x + 3)(x - 3)$ 2) $(x - 1)(x + 2)(x - 3)(x + 3)$

3) $(x + 1)(x - 2)(x + 3)(x + 2)$ 4) $(x + 1)(x - 1)(x + 2)(x + 3)$

17. By factor theorem, the factors of $x^4 + 2x^3 - 13x^2 - 14x + 24$ are 1) $(x - 1)(x + 2)(x - 3)(x + 4)$ 2) $(x + 1)(x - 2)(x - 3)(x + 4)$

3) $(x - 1)(x + 2)(x + 3)(x - 4)$ 4) $(x - 1)(x - 2)(x - 3)(x + 4)$

18. If A, B, C are the remainders of $x^3 - 3x^2 - x + 5$, $3x^4 - x^3 + 2x^2 - 2x - 4$, $2x^5 - 3x^4 + 5x^3 - 7x^2 + 3x - 4$, when divided by $(x + 1)$, $(x + 2)$, $(x - 2)$ respectively, then the ascending order of A, B, C is

1) A, B, C 2) B, C, A 3) A, C, B 4) B, A, C

19. If the quotient of $x^4 - 11x^3 + 44x^2 - 76x + 48$, when divided by $x^2 - 7x + 12$ is $Ax^2 + Bx + C$, then the descending order of A, B, C is

1) A, B, C 2) B, C, A 3) A, C, B 4) C, A, B

20. If $(x + 1)$, $(x - 1)$, $(x - 3)$ are the factors of $x^3 + Ax^2 + Bx + C$, then the ascending order of A, B, C is

1) A, B, C 2) B, C, A 3) A, C, B 4) B, A, C

21. If $x - 2$ and $x + 3$ are both factors of the expression $x^3 + ax^2 + bx - 12$, then

1) $a = 4, b = 3$ 2) $a = 3, b = -4$ 3) $a = -3, b = -4$ 4) None

22. The remainder when x^{1999} is divided by $x^2 - 1$ is

1) x 2) x^2 3) $x^2 + 1$ 4) $x + 1$

23. The remainder contains no term in x^2, when $x^5 + kx^2$ is divided by $(x-1)(x-2)(x-3)$, then the value of k is

1) 90 2) 80 3) –80 4) –90

24. If $f(x)$ is divided by $(x-1), (x-2)$ gives the remainders 5 and 7 respectively, then the remainder when same $f(x)$ is divided by $(x-1)(x-2)$ is

1) $2x+5$ 2) $2x-5$ 3) $2x-3$ 4) $2x+3$